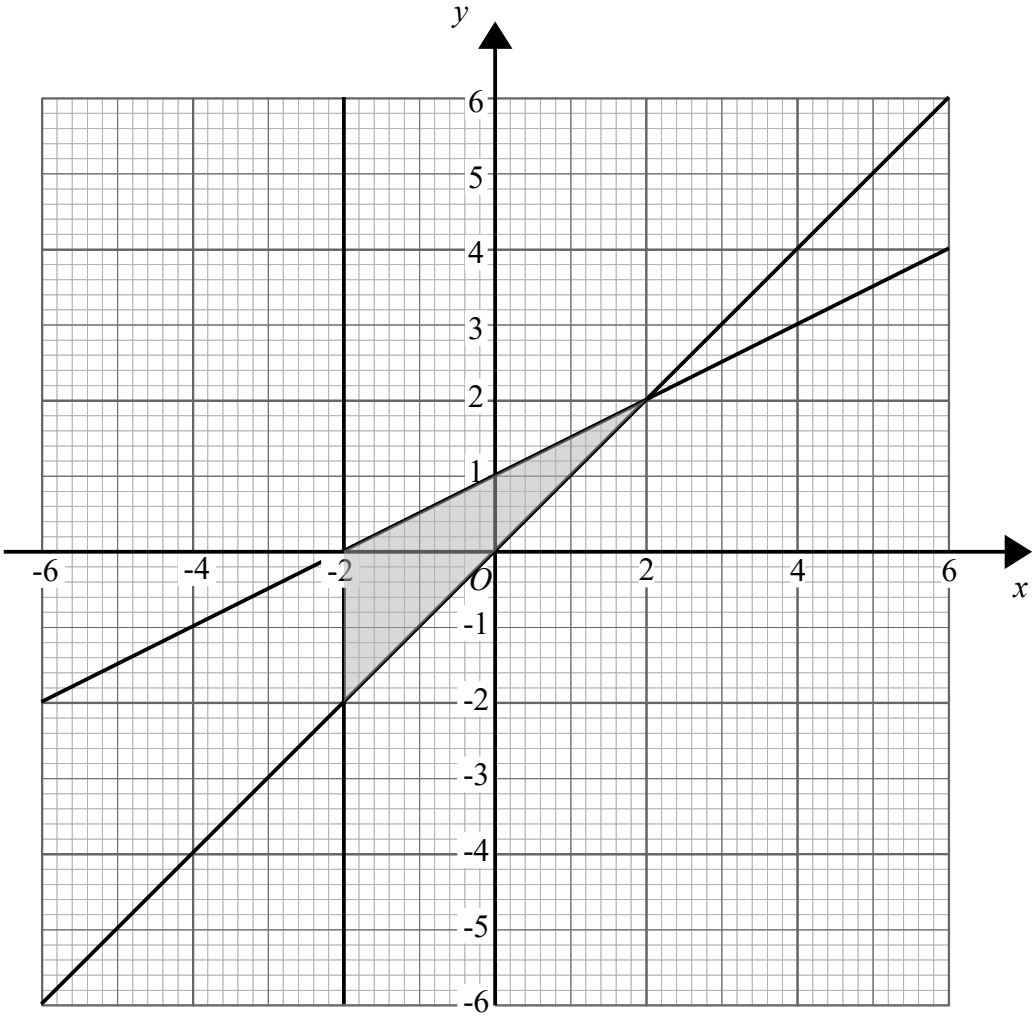
1



Write down the three inequalities that define the shaded region

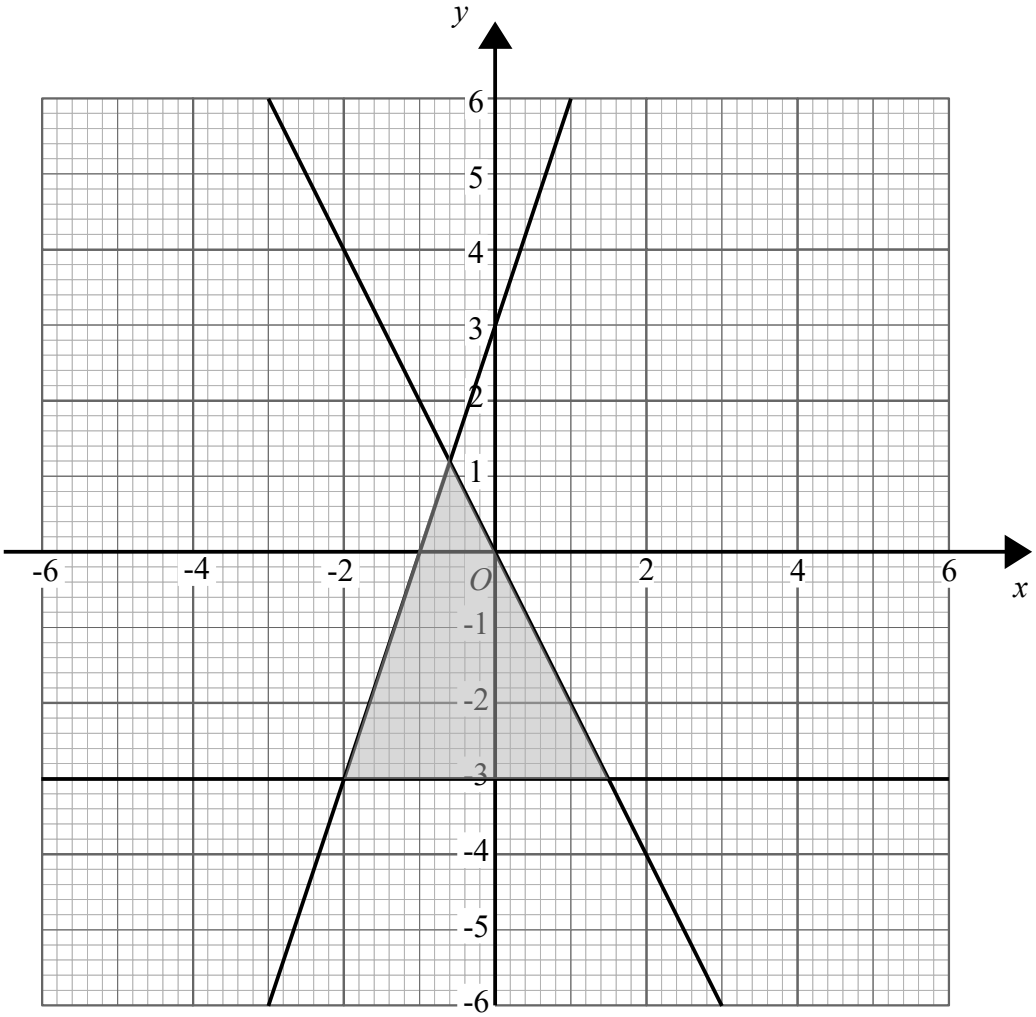
.....

.....

.....

(Total for Question 1 is 4 marks)

2



Write down the three inequalities that define the shaded region

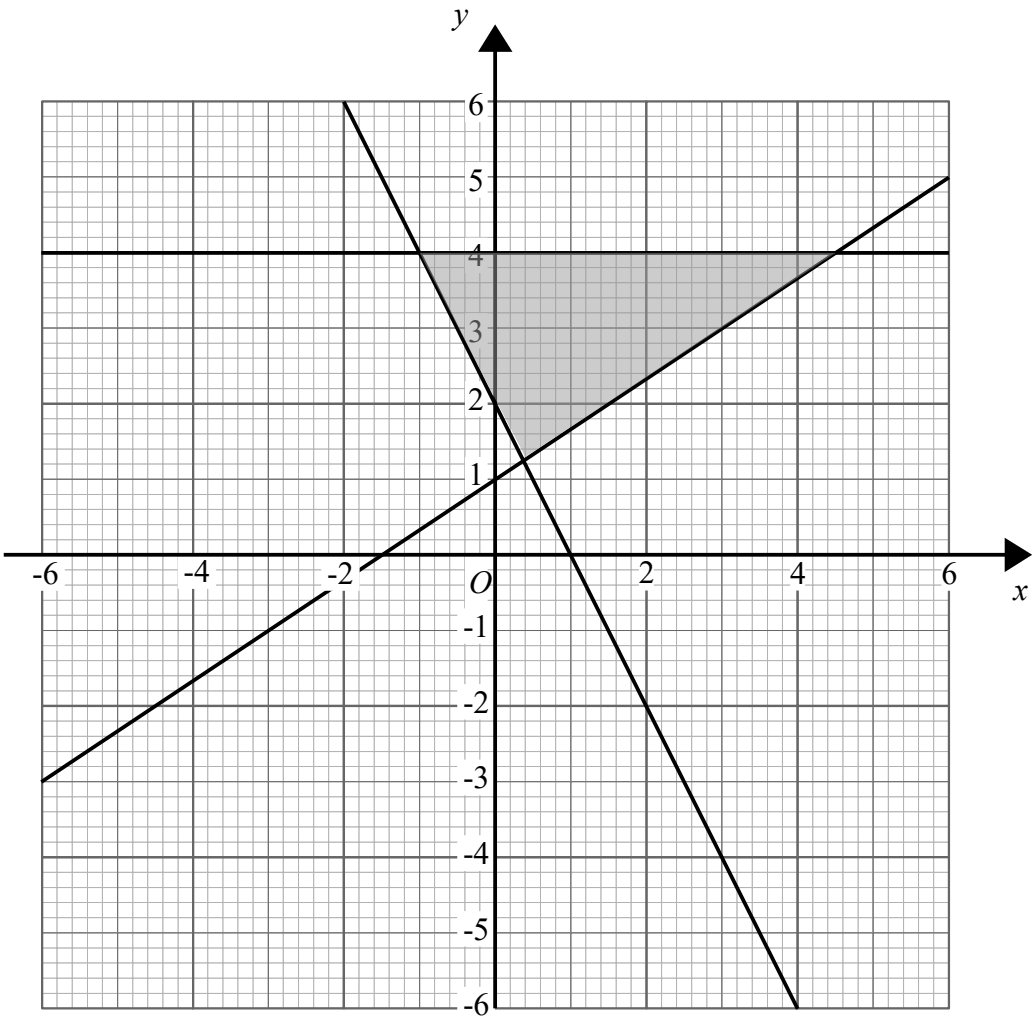
.....

.....

.....

(Total for Question 2 is 4 marks)

3



Write down the three inequalities that define the shaded region

.....

.....

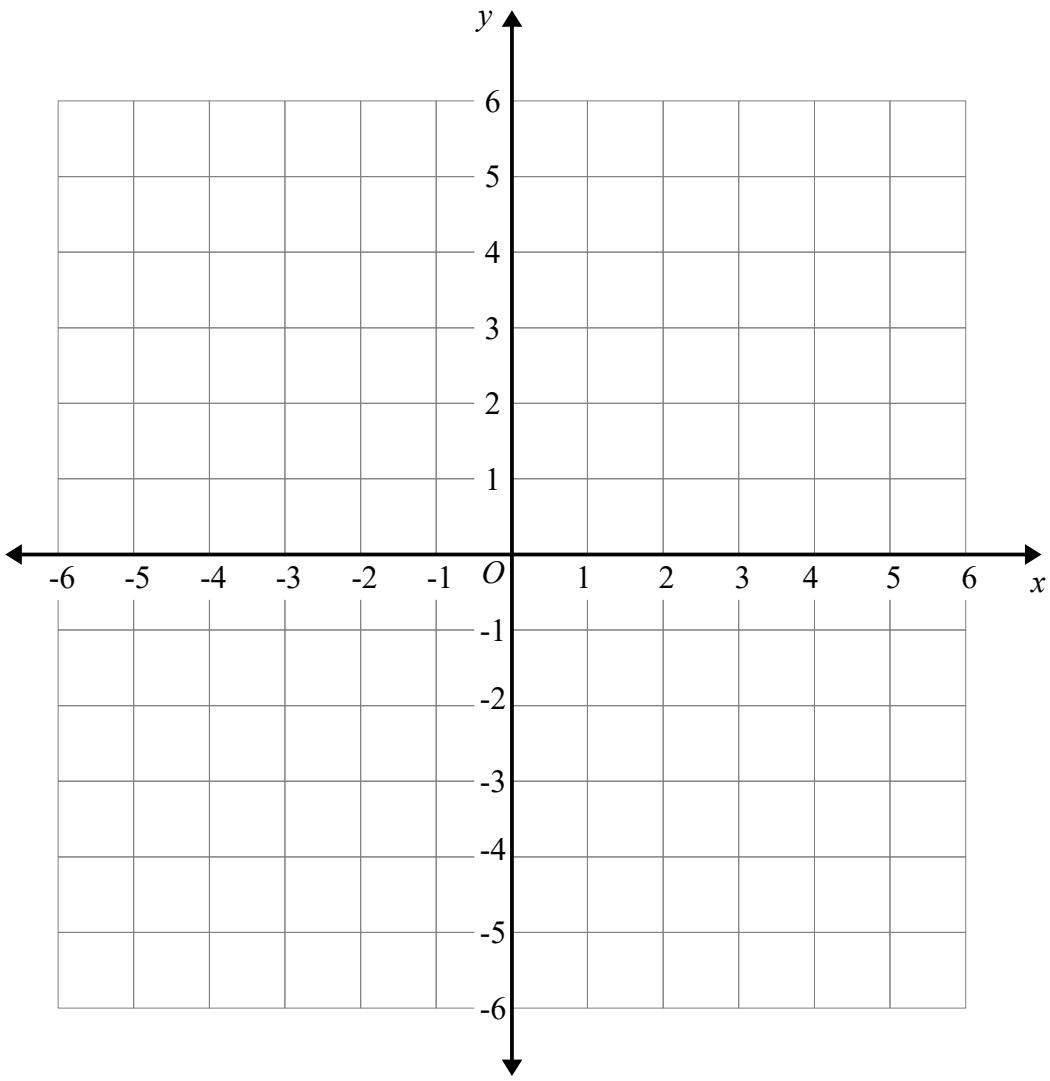
.....

(Total for Question 3 is 4 marks)

4 On the grid shade the region that satisfies all these inequalities

$$x < 3 \qquad y > -2 \qquad y \leq 2x - 1$$

Label the region **R**.

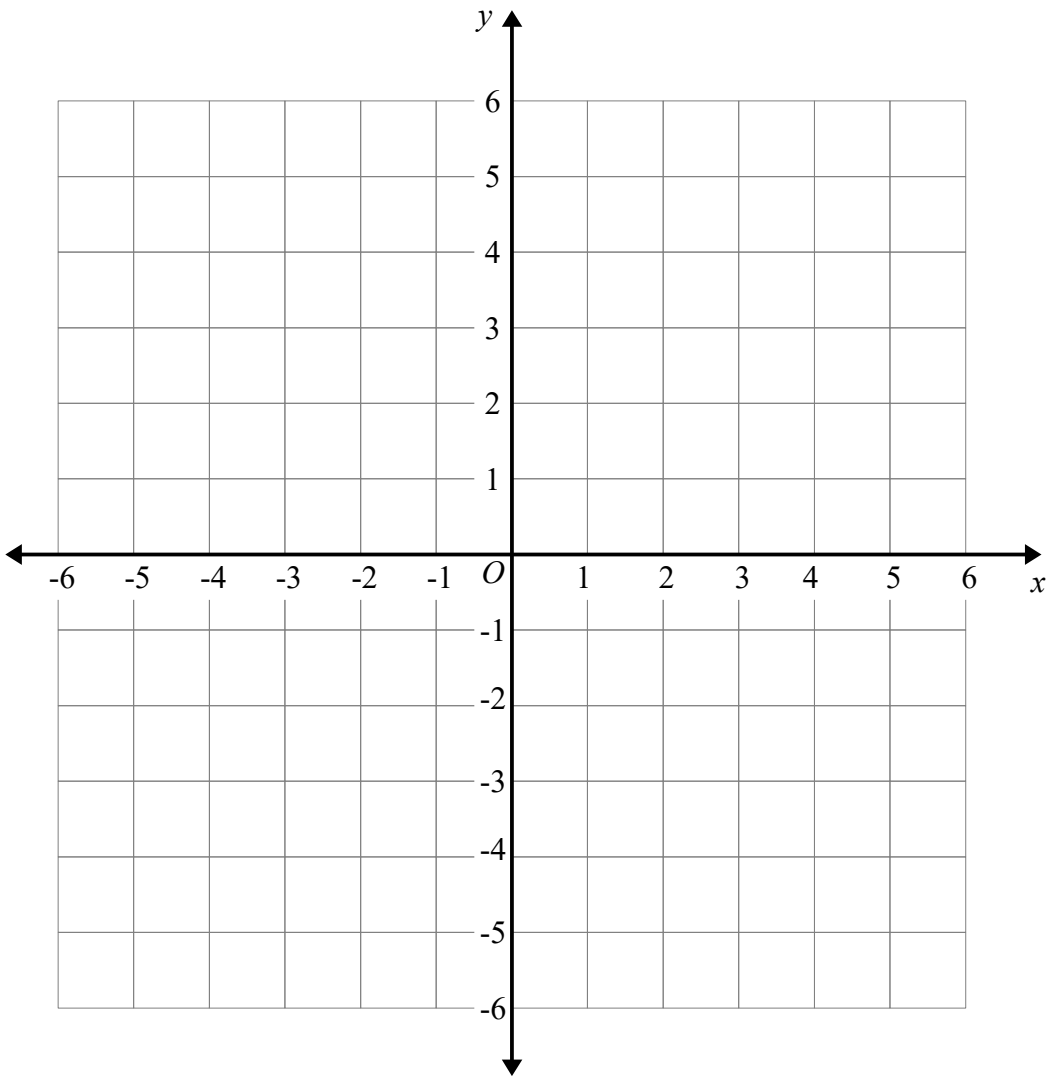


(Total for Question 4 is 3 marks)

5 On the grid shade the region that satisfies all these inequalities

$x + y < 4$ $\qquad$ $y > 2x + 1$ $\qquad$ $y > -1$

Label the region **R**.

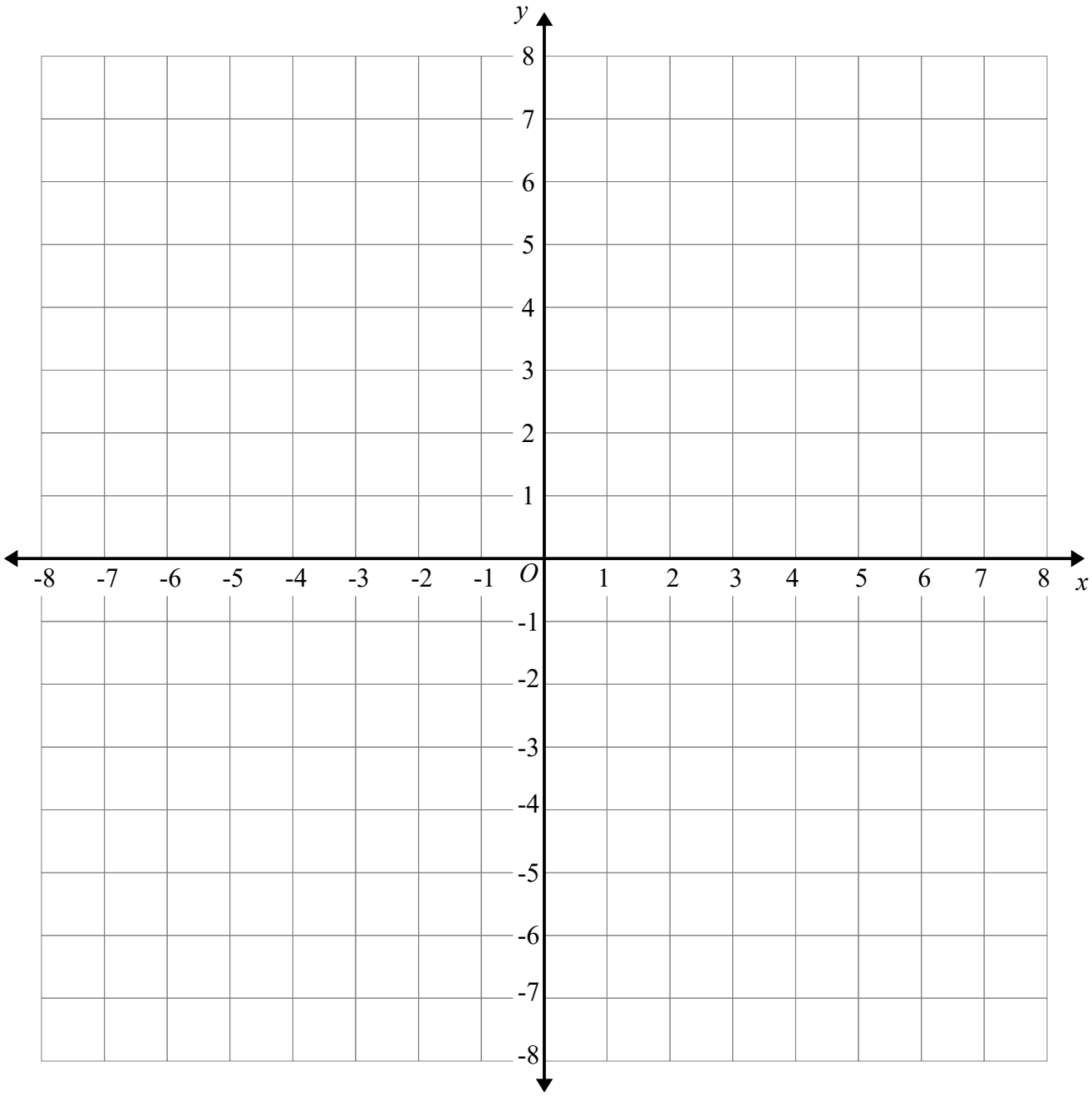


(Total for Question 5 is 3 marks)

6 On the grid shade the region that satisfies all these inequalities

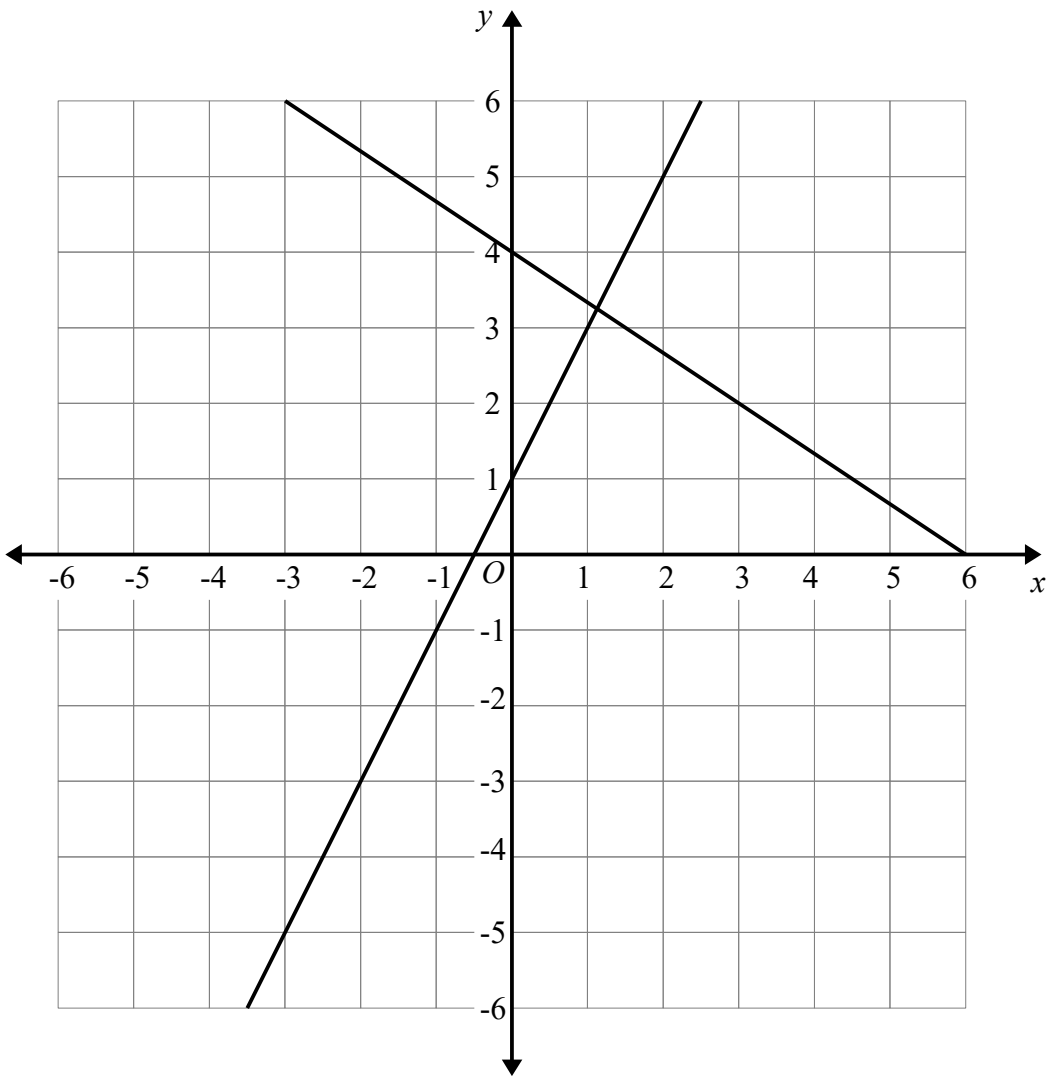
$$y \geq x - 1 \qquad x \leq 6 - 2y \qquad x \geq -3$$

Label the region **R**.



(Total for Question 6 is 3 marks)

7 The graphs of the straight lines with equations $y = 2x + 1$ and $3y + 2x = 12$ have been drawn on the grid.



x and y are both integers.

Mark with a cross (×) all of the points that satisfies all the inequalities

$y < 2x + 1$ $\qquad$ $3y + 2x < 12$ $\qquad$ $y > 1$

(Total for Question 7 is 2 marks)